

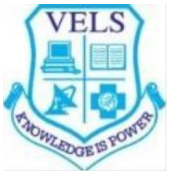


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SMART EMERGENCY VEHICLE PRIORITY SYSTEM

FINAL YEAR

PROJECT PHASE-I REPORT

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in partial fulfilment for the award of the degree of

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BONAFIDE CERTIFICATE

Certified that this project report titled “Smart Emergency Vehicle Priority System” is the bonafide work of Rohith S. (24619164), Jeremiel Joseph S. (24619115), and Guru Rahul R. (22619187), who carried out the project work under the guidance of Ms. Shanthi, Department of Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai.

Certified further that, to the best of our knowledge, the work reported here has been prepared for the project phase and has not been submitted in full for any other degree or award.

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Submitted for the Project Work and Viva-Voce examination held on _____ at VISTAS, Chennai.

INTERNAL EXAMINER	EXTERNAL EXAMINER
Signature: _____	Signature: _____

DECLARATION

We hereby declare that the project work entitled “Smart Emergency Vehicle Priority System” submitted to Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai, in partial fulfilment of the requirements for the award of the Bachelor of Technology degree in Computer Science and Engineering, is a record of the work carried out by us under the guidance of Ms. Shanthi. The report describes the proposed system and prototype plan based on the project materials prepared by our team. We further declare that this report has not been submitted previously, in part or in full, for the award of any degree or diploma at another institution.

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TABLE OF CONTENTS

ABSTRACT 7
LIST OF FIGURES 8
LIST OF TABLES 9
LIST OF ABBREVIATIONS 10
CHAPTER 1 — INTRODUCTION 11
1.1 General Introduction 11
1.2 Background and Motivation 11
1.3 Problem Description 12
1.4 Existing System 12
1.5 Limitations of Existing System 12
1.6 Proposed System 13
1.7 Objectives 13
1.8 Scope 13
1.9 Major Contributions 14
1.10 System Modules 14
1.11 Report Organization 14
CHAPTER 2 — LITERATURE REVIEW 15
2.1 Emergency Vehicle Preemption 15
2.2 IoT-Based Traffic Management 15
2.3 Wireless Communication for Junctions 16
2.4 Research Gap 16
CHAPTER 3 — SYSTEM DESIGN AND ARCHITECTURE 17
3.1 System Overview 17
3.2 Existing-System Architecture 17
3.3 Proposed Architecture 18
3.4 Workflow 19
3.5 Functional Modules 20
3.6 Message Structure 21
3.7 Safety and Restoration 21
3.8 Hardware Requirements 22
3.9 Software Requirements 22
CHAPTER 4 — IMPLEMENTATION AND METHODOLOGY 23
4.1 Development Environment 23
4.2 Prototype Setup 23
4.3 Emergency Detection 24
4.4 Message Processing and LoRa 24
4.5 Signal Priority Logic 25
4.6 Cascading Junction Coordination 25
4.7 Testing Methodology 26
CHAPTER 5 — RESULTS AND DISCUSSION 27

5.1 Prototype Demonstration Plan....27
5.2 Expected Behaviour....27
5.3 Evaluation Metrics....28
5.4 Limitations....28
CHAPTER 6 — APPLICATIONS AND DEMONSTRATION....29
6.1 Application Areas....29
6.2 Demonstration Sequence....29
6.3 Illustrative Three-Junction Model Layout....29
CHAPTER 7 — CONCLUSION AND FUTURE WORK....30
7.1 Conclusion....30
7.2 Objectives and Status....30
7.3 Future Enhancements....31
REFERENCES....32
APPENDIX — TEST PLAN AND SAMPLE MESSAGE....33

ABSTRACT

Emergency vehicles such as ambulances and fire engines can be delayed by congestion and repeated traffic-signal stops. A delay at one junction may be followed by additional delays at the next junction, increasing the time needed to reach an emergency location. This project proposes a Smart Emergency Vehicle Priority System that uses ESP32 microcontrollers and LoRa wireless communication to coordinate priority across consecutive traffic-signal junctions.

When an emergency request is activated at a junction, the local ESP32 processes the request, initiates a safe priority sequence for the emergency vehicle, and transmits the emergency information to the next junction. The receiving junction can prepare its signal before the vehicle arrives. This cascading approach aims to reduce waiting time, improve coordination between junctions, and return signals to normal operation after the vehicle has passed. The proposed prototype uses traffic-signal LEDs or a traffic module, ESP32 boards, LoRa modules, and optional GPS-based location information.

The project focuses on demonstrating emergency request handling, wireless message transmission, junction-to-junction forwarding, safe signal prioritisation, and restoration of normal traffic operation. Quantitative performance claims are not made in this report because measured test results were not supplied in the source project materials. Future work can add GPS route tracking, traffic-density sensing, authenticated messages, cloud monitoring, and larger-scale field evaluation.

Keywords: Emergency Vehicle Priority, ESP32, LoRa, Internet of Things (IoT), Traffic Signal Control, Cascading Junction Network, Smart Traffic Management.

LIST OF FIGURES

Figure No.	Title
Figure 3.1	Proposed Smart Emergency Vehicle Priority System Architecture
Figure 3.2	Emergency Vehicle Priority Workflow
Figure 4.1	Illustrative prototype arrangement of junction nodes (to be added after hardware setup)

LIST OF TABLES

Table No.	Title
Table 1.1	Project Objectives and Prototype Scope
Table 2.1	Summary of Existing Approaches and Limitations
Table 3.1	Emergency Message Fields
Table 3.2	Hardware Components
Table 3.3	Software Requirements
Table 4.1	Suggested Functional Test Cases
Table 5.1	Evaluation Metrics to Record
Table 6.1	Potential Application Areas

LIST OF ABBREVIATIONS

Abbreviation	Full Form / Meaning
AI	Artificial Intelligence
ESP32	Espressif 32-bit microcontroller family
GPS	Global Positioning System
IoT	Internet of Things
LED	Light-Emitting Diode
LoRa	Long Range low-power wireless communication technology
MCU	Microcontroller Unit
RF	Radio Frequency
UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus
VISTAS	Vels Institute of Science, Technology and Advanced Studies

CHAPTER 1 — INTRODUCTION

1.1 General Introduction

Emergency response depends on reaching the incident location as quickly and safely as possible. Ambulances, fire engines, and other authorised emergency vehicles can lose valuable time when they encounter dense traffic or red signals. A conventional signal controller generally follows its programmed timing plan and may not know that an emergency vehicle is approaching. Even when priority is provided at the current junction, the next junction may not receive advance information.

The Smart Emergency Vehicle Priority System is designed to connect multiple junction nodes. Each node uses an ESP32 microcontroller to process an emergency request and a LoRa module to communicate with other nodes. The current junction can prioritise the emergency route, while the next junction receives a message and prepares for the vehicle's arrival. This is a cascading approach because the request can move forward along a configured sequence of junctions.

1.2 Background and Motivation

Emergency Vehicle Preemption (EVP) systems can change a traffic signal to favour an emergency vehicle. However, the benefit may be limited if the system operates only at one intersection or reacts only when the vehicle reaches a detection zone. A network that shares the request between adjacent junctions may reduce repeated stops by preparing signals ahead of time.

The project is motivated by the need for a low-cost, educational prototype that demonstrates communication and control using widely available embedded hardware. ESP32 provides processing and GPIO interfaces, while LoRa is suitable for low-rate messages over longer distances than many short-range wireless links, subject to local radio regulations and deployment conditions.

1.3 Problem Description

The central problem is the lack of advance coordination between consecutive traffic junctions when an emergency vehicle is travelling along a route. A vehicle may receive priority at one intersection but encounter a red signal at the next. Manual intervention or independent local detection can result in inconsistent timing and additional waiting.

- Emergency vehicles may experience delay at signalised intersections.
- An upcoming junction may not know about the approaching emergency vehicle.
- Independent junction decisions can lead to fragmented priority along a route.
- Wireless communication failure or stale messages must not leave signals in an unsafe state.
- The system needs a controlled return to normal traffic after the emergency vehicle passes.

1.4 Existing System

In a conventional traffic system, each junction follows a fixed-time or traffic-actuated signal plan. Some emergency vehicle preemption solutions use dedicated transmitters, roadside receivers, or central traffic-management infrastructure. These solutions can provide priority, but their capabilities depend on the detection method, coverage, infrastructure, and whether neighbouring junctions share information.

1.5 Limitations of the Existing System

- Local-only response: the controller may act only after the vehicle reaches the junction.
- Limited advance notice: the next intersection may not receive the emergency request early enough.
- Fragmented coordination: multiple junctions may handle the same trip independently.
- Deployment cost: specialised infrastructure may be difficult to deploy in a small prototype.
- Operational safety: priority must be granted without conflicting green indications or unsafe transitions.

1.6 Proposed System

The proposed system uses an ESP32 node and LoRa communication at each participating junction. An emergency request is activated by a switch or a configured detection input. The ESP32 validates the request and prepares a message containing identifiers, direction, current junction, next junction, and priority status. The local controller starts a safe priority sequence for the emergency approach. The message is transmitted to the next junction, where the receiver checks the message and prepares the signal in advance.

After the emergency vehicle passes, a clear or timeout condition releases the priority state and restores the normal traffic sequence. The prototype is intended to demonstrate the control logic using LEDs or a traffic-signal module. GPS may be added for location and route information, but it is optional in the current concept.

1.7 Objectives of the Project

1. Design a low-cost emergency vehicle priority prototype using ESP32.
2. Transmit emergency information between junction nodes using LoRa.
3. Provide a priority phase at the current junction while respecting safe signal transitions.
4. Forward emergency information to upcoming junctions for advance preparation.
5. Restore normal traffic operation after the emergency vehicle passes or the request is safely cleared.
6. Evaluate message delivery, signal response, junction coordination, and recovery behaviour.

Objective	Prototype scope
Emergency request detection	Emergency switch / detection input handled by ESP32
Wireless coordination	LoRa message transmission and reception between nodes
Priority control	LED/traffic module demonstrates a safe priority sequence
Cascading operation	Next junction receives and prepares from forwarded information
Restoration	Clear request or controlled timeout returns system to normal
Evaluation	Record response time, packet delivery, and successful state transitions

1.8 Scope of the Project

The scope is a prototype demonstration rather than direct control of public-road traffic infrastructure. The system can be tested with two or three miniature junction nodes, low-voltage signal LEDs, and wireless modules. It does not establish that the system is ready for deployment on live roads. Real-world use would require certified hardware, fail-safe traffic-controller integration, radio compliance, cybersecurity, field testing, and approval from the relevant authorities.

1.9 Major Contributions

- A cascading junction-to-junction communication concept for emergency requests.
- An embedded implementation approach based on ESP32 and LoRa.
- A signal-state workflow that includes priority, vehicle passage, and normal restoration.
- A modular test plan for evaluating wireless reliability and response behaviour.

- A scalable concept that can later incorporate GPS, traffic-density data, and monitoring services.

1.10 System Modules — Overview

The proposed design contains five functional modules: emergency request input, message generation and validation, LoRa communication, junction signal control, and clearance/restoration. Each junction can run the same software with a unique junction identifier and route configuration.

1.11 Organization of the Report

Chapter 2 reviews relevant approaches to emergency vehicle preemption, IoT traffic management, and wireless communication. Chapter 3 describes the architecture and workflow. Chapter 4 explains the prototype methodology and test plan. Chapter 5 defines how results should be measured and discusses the current evidence and limitations. Chapter 6 presents possible applications and a demonstration sequence. Chapter 7 concludes the report and identifies future enhancements.

CHAPTER 2 — LITERATURE REVIEW

2.1 Emergency Vehicle Preemption

Emergency Vehicle Preemption gives an authorised emergency vehicle temporary priority at a traffic signal. Implementations may rely on optical emitters, radio transmitters, vehicle-to-infrastructure communication, GPS-based fleet systems, or central traffic-management platforms. The core requirement is to recognise a legitimate request and transition the signal safely so that conflicting traffic is stopped before the emergency movement receives priority.

A key design consideration is that priority is not the same as immediately switching every signal to green. The controller must respect yellow and all-red clearance intervals, prevent conflicting greens, and account for pedestrians and other road users. A student prototype should demonstrate the state transitions using low-voltage LEDs rather than connect directly to public-road signal equipment.

2.2 IoT-Based Traffic Management

IoT traffic systems use embedded controllers, sensors, and communication links to exchange information between vehicles, roadside devices, and monitoring applications. An ESP32-based node can process digital inputs and control a small signal model. A network of nodes can represent consecutive intersections and allow a request detected at one point to be shared with downstream nodes.

2.3 Wireless Communication for Junctions

LoRa is a low-power, long-range wireless technology that can carry compact messages between nodes. Emergency requests are small compared with video or other high-bandwidth data, making a low-data-rate link a reasonable prototype choice. Actual coverage depends on antenna placement, buildings, interference, frequency band, transmit power, and local regulations. The design should include message identifiers, duplicate handling, acknowledgement or retry logic where appropriate, and timeout behaviour.

2.4 Cascading Priority and Coordination

Cascading priority means that the current junction does not keep the request local: it forwards relevant information to the next configured junction. The next node can prepare before the vehicle arrives. For correct operation, the message should include a route direction and next-junction identifier. A node should avoid forwarding the same message indefinitely and should reject expired or duplicate requests.

2.5 Existing Approaches and Limitations

Approach	Strength	Limitation relevant to this project
Fixed-time signal control	Simple and predictable	Does not inherently know an emergency vehicle is approaching
Local emergency preemption	Can prioritise one junction	May not prepare downstream junctions
Centralised traffic management	Can coordinate a larger network	May depend on infrastructure, network connectivity, and integration
Short-range wireless prototype	Easy to demonstrate over a small area	Range may be limited by obstacles and deployment
ESP32 + LoRa cascading prototype	Low-cost educational demonstration of node-to-node	Requires careful routing, validation, timing, and fail-safe

	forwarding	behaviour

2.6 Research Gap

The project addresses the practical gap between providing priority at one junction and coordinating priority across several consecutive junctions. The intended contribution is an integrated prototype that detects an emergency request, controls the current signal, forwards a compact message over LoRa, prepares the next junction, and restores normal operation. The project materials do not provide a measured comparison against a deployed traffic system; therefore, claims of actual travel-time improvement require future testing.

2.7 Chapter Summary

This chapter described emergency preemption, IoT traffic management, LoRa communication, and cascading coordination. It highlighted the need for safe transitions, valid messages, duplicate prevention, and controlled restoration. The next chapter presents the system architecture.

CHAPTER 3 — SYSTEM DESIGN AND ARCHITECTURE

3.1 System Overview

The system consists of junction nodes connected by LoRa. Each node contains an ESP32, a LoRa transceiver, and a miniature traffic-signal output. An emergency request is introduced at the first node. The local ESP32 checks the request, updates the signal state, and transmits a structured message. A downstream node receives the message, verifies it, and prepares the appropriate signal phase.

3.2 Existing-System Architecture

In a conventional arrangement, each signal controller operates independently according to its timing plan or local detection. An emergency vehicle may need to request priority separately at every junction. If the next intersection is not informed early, the emergency vehicle may stop again even after receiving priority at the previous signal.

3.3 Proposed-System Architecture

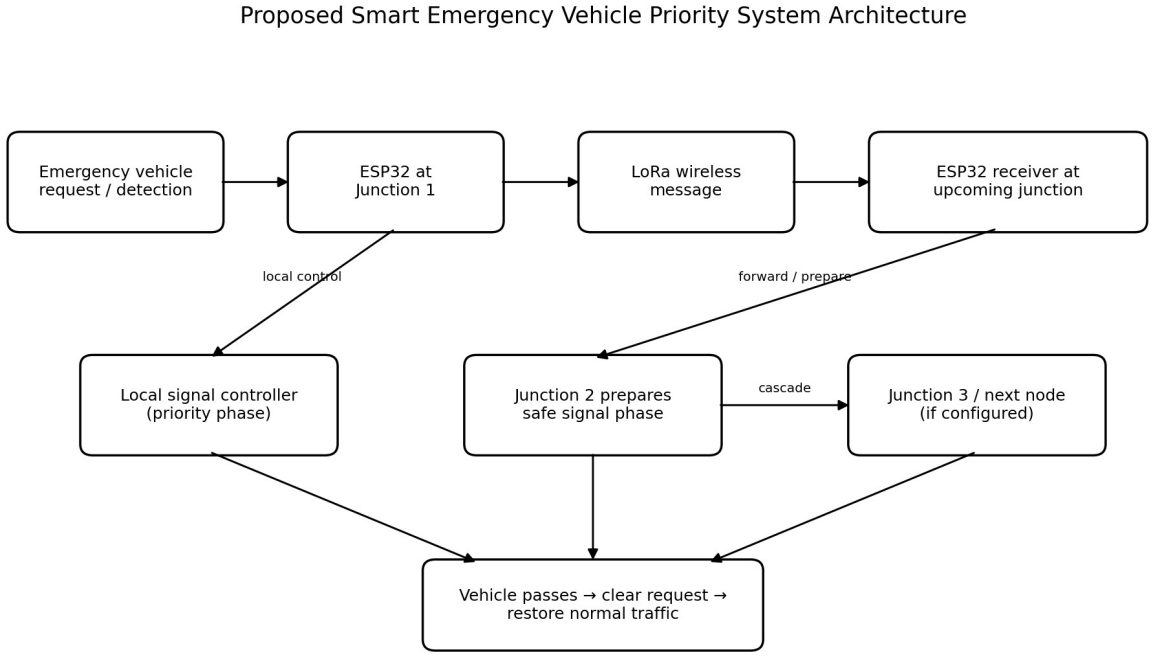


Figure 3.1: Proposed Smart Emergency Vehicle Priority System Architecture

The architecture has two related paths. The local control path activates priority at the current junction. The communication path transmits the emergency request to the next configured junction. The receiver checks the message and prepares its signal, and the request may be forwarded again if another junction is configured.

3.4 Overall System Workflow

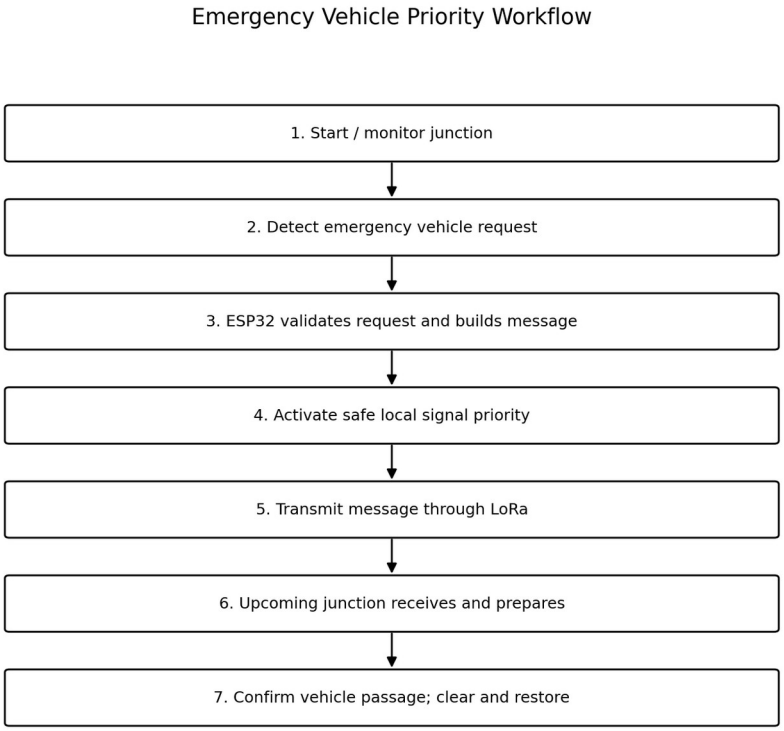


Figure 3.2: Emergency Vehicle Priority Workflow

The workflow begins in the normal monitoring state. When a request is detected, the node validates the input and creates a message. The local signal controller follows a safe transition into the priority state, while LoRa transmits the request. The next node checks the message and prepares its signal. Once passage is confirmed, the system clears the request and returns to normal operation.

3.5 Functional Modules

3.5.1 Emergency Vehicle Detection

For a basic prototype, a push button or switch can represent an emergency request. A later version can use a dedicated vehicle transmitter, GPS geofence, or authenticated detection device. A manual switch is useful for demonstrating the logic but does not automatically identify an authorised emergency vehicle.

3.5.2 ESP32 Processing

The ESP32 reads the request input, checks whether a request is already active, assigns the current junction identifier, selects the next junction, and prepares a message. The controller should use a defined state machine rather than arbitrary delays wherever practical. The state machine should prevent conflicting signal indications and handle invalid or repeated requests.

3.5.3 LoRa Transmission and Reception

The LoRa transmitter sends the compact emergency message to the next node. The receiver verifies the message fields, checks its sequence number and expiry time if implemented, and ignores duplicates or stale messages. An acknowledgement can be used to confirm receipt. If acknowledgement is not received, a bounded retry policy may be applied; retries should not create repeated or indefinite priority.

3.5.4 Traffic Signal Priority

The signal module represents each traffic approach using LEDs. Before the emergency direction receives green, conflicting movements must transition through the required clearance states. In a miniature model,

the state sequence should be clearly visible and logged. The project must not be connected directly to live public-road signal controllers without an approved, certified interface.

3.5.5 Cascading Junction Preparation

The receiving node uses the message's next-junction and direction fields to determine whether the request is intended for it. If valid, it marks the request as pending and prepares the correct phase. If the route continues, the node forwards an updated message with its own junction identifier and the next destination.

3.5.6 Clearance and Normal Restoration

The priority state should be cleared by a passage sensor, an operator clear command, or a carefully bounded timeout used only for the prototype. When the request is cleared, the controller transitions safely to the normal timing sequence. A lost radio message or a controller reset should not leave an uncontrolled priority state active indefinitely.

3.6 Emergency Message Structure

Field	Purpose	Example (illustrative)
message_id	Unique request identifier	EV-0007
vehicle_type	Type of emergency vehicle	AMBULANCE
emergency_status	Active or clear state	ACTIVE
current_junction	Node that generated/forwarded message	J1
next_junction	Intended receiving node	J2
direction	Emergency travel direction	NORTH
priority	Requested priority level	HIGH
sequence_number	Helps reject duplicate/out-of-order messages	12
timestamp / expiry	Helps reject stale messages	Prototype-defined

The example values are illustrative only. The actual packet format should be selected during implementation and kept consistent across all ESP32 nodes.

3.7 Safety and Restoration

- Never permit conflicting approaches to show green simultaneously.
- Use a defined transition sequence, including clearance intervals appropriate to the prototype.
- Validate message structure and destination before changing signal state.
- Use message IDs and sequence numbers to prevent duplicate requests.
- Provide a clear/expiry mechanism and a safe default state after communication loss or reset.
- Record request, receive, priority, clear, and restore events for evaluation.

3.8 Hardware Requirements

Hardware component	Role
ESP32 development board	Processes request input and controls signal outputs
LoRa transceiver module	Sends and receives junction messages
Traffic signal LEDs / traffic module	Demonstrates red, yellow, and green signal states

Push button or switch	Represents emergency vehicle detection/request
Breadboard and jumper wires	Prototype interconnections
USB power / regulated supply	Powers low-voltage prototype hardware
Optional GPS module	Can provide location or route information in a later version

3.9 Software Requirements

Software / tool	**Purpose**
Arduino IDE	Firmware development and upload
Embedded C/C++	ESP32 control logic
ESP32 board support package	Board compilation and upload support
LoRa library compatible with selected module	Radio packet transmission and reception
Serial Monitor	Debugging, status messages, and timing logs
Optional simulator	Early testing of logic before hardware assembly

3.10 Chapter Summary

This chapter described the system overview, architecture, workflow, modules, message fields, safety rules, and hardware/software requirements. The next chapter explains how the prototype can be assembled and tested.

CHAPTER 4 — IMPLEMENTATION AND METHODOLOGY

4.1 Development Environment

The project presentation identifies Arduino IDE, embedded C/C++, ESP32 board support, a simulator, and Serial Monitor as development tools. The firmware should be developed and tested incrementally: first verify each signal output, then the LoRa link, and finally the complete multi-junction sequence.

4.2 Hardware Prototype Setup

Prepare two or three low-voltage junction models. Each node should have an ESP32 and a compatible LoRa module. Connect red, yellow, and green LEDs through suitable current-limiting resistors to the chosen GPIO pins. Connect an emergency request switch to a digital input using an appropriate pull-up or pull-down configuration. Confirm the selected LoRa module's supply voltage and wiring requirements before powering it.

The precise GPIO numbers and radio wiring depend on the ESP32 board and LoRa module selected. These values must be recorded from the actual hardware datasheets and the final wiring diagram; they are not specified in the source project materials.

4.3 Emergency Detection

The detection routine monitors the request input. When a new request is registered, the firmware applies input debouncing and checks whether an emergency request is already active. It then creates a unique request identifier and changes the node state from NORMAL to REQUEST_VALIDATION and then to PRIORITY_PENDING, according to the implemented state machine.

4.4 Message Processing and LoRa Communication

The message-generation routine fills the required fields, including vehicle type, emergency status, current junction, next junction, direction, and priority. The transmitter sends the packet through LoRa. The receiving node parses the packet and checks its destination, message identifier, sequence number, and validity. Invalid or expired packets should be logged and ignored.

4.5 Signal Priority Logic

A recommended simplified state machine is NORMAL → REQUEST_VALIDATION → CLEARANCE → EMERGENCY_GREEN → VEHICLE_PASSAGE / CLEAR → RESTORE_NORMAL. The exact sequence must reflect the miniature signal design and should be reviewed for safety. A request should not bypass a conflicting phase's required clearance.

State	Action	Exit condition
NORMAL	Run ordinary LED timing sequence	New valid request
REQUEST_VALIDATION	Debounce and validate request	Valid / invalid decision
CLEARANCE	Show safe transition state	Clearance interval completed
EMERGENCY_GREEN	Provide priority for emergency direction	Passage or clear condition
FORWARD_REQUEST	Send message to next junction	Transmission attempt completed
RESTORE_NORMAL	Return to configured normal cycle	Normal state reached

4.6 Cascading Junction Coordination

After local request validation, Junction 1 transmits the request for Junction 2. Junction 2 prepares the correct approach and can forward an updated request to Junction 3 if configured. The implementation should maintain the original message ID across the journey while incrementing a hop or sequence field. A maximum hop count or explicit route list can prevent looping.

4.7 Testing Methodology

Testing should proceed from unit tests to integrated tests. First verify each LED state and button input. Next verify transmission and reception between two LoRa nodes. Then test request forwarding across multiple nodes, including duplicate messages, invalid destinations, lost packets, reset events, and request clearance. Repeat each scenario and record observed times rather than estimating them.

Test ID	Test case	Expected outcome
T1	No emergency request	All junctions remain in normal signal cycle
T2	Request at Junction 1	Junction 1 validates request and enters safe priority sequence
T3	Valid LoRa packet to Junction 2	Junction 2 receives and prepares intended phase
T4	Request forwarded to Junction 3	Junction 3 prepares if it is on configured route
T5	Duplicate packet	Duplicate is ignored or handled idempotently
T6	Invalid destination / malformed packet	Packet is rejected and logged
T7	Communication loss	System follows defined safe timeout/default behaviour
T8	Vehicle passage / clear command	Priority is cleared and normal cycle resumes
T9	ESP32 reset during request	Node starts in a safe known state and recovers predictably

4.8 Chapter Summary

This chapter outlined a staged implementation method, the expected state machine, message handling, cascade coordination, and a functional test plan. Hardware pin assignments and radio configuration must be completed using the selected components.

CHAPTER 5 — RESULTS AND DISCUSSION

5.1 Prototype Demonstration Plan

The project source materials describe the intended prototype using ESP32, LoRa, traffic-signal modules, and optional GPS. They do not include measured communication range, packet delivery rate, response-time logs, or a completed results table. Therefore, this report records the intended observable behaviour and provides metrics to fill in after testing; it does not invent experimental results.

5.2 Expected Behaviour

During a successful demonstration, a request at the first junction should cause the local controller to enter a safe priority sequence. A message should be sent over LoRa, and the next junction should receive the request and prepare the corresponding phase before a simulated vehicle arrives. After the passage event or a valid clear command, the nodes should restore normal traffic operation.

5.3 Evaluation Metrics

Metric	How to measure	Why it matters
Request-to-priority latency	Time between request activation and priority state	Measures local responsiveness
Wireless delivery rate	Successfully received valid packets / packets sent	Measures communication reliability
End-to-end forwarding latency	Time from initial request to downstream-node receipt	Measures cascading speed
Duplicate rejection rate	Duplicate packets correctly ignored / duplicates injected	Checks idempotent handling
Restoration time	Time from clear event to normal state	Measures recovery behaviour
Successful test-case rate	Passed test cases / total test cases	Summarises functional completeness

5.4 Results Table to Complete After Testing

Test / Metric	Observed value	Status / Notes
Request-to-priority latency	To be measured	Fill from Serial Monitor / test log
LoRa packet delivery rate	To be measured	Record sent and received packet counts
Junction 1 → Junction 2 latency	To be measured	Repeat under the same conditions
Junction 2 → Junction 3 latency	To be measured	Only if three nodes are assembled
Normal restoration success	To be measured	Count successful clear/restore trials
Total functional tests passed	To be measured	Use Table 4.1 test cases

5.5 Discussion

The key design advantage is that the emergency request is not limited to a single junction. Downstream nodes can prepare based on the received message. The quality of this benefit depends on reliable message delivery, correct route configuration, timely clearing, and safe signal transitions. If messages are lost or stale, the implementation must fail safely rather than leave an inappropriate priority phase active.

5.6 Limitations and Threats to Validity

- The report is based on the supplied project presentation and abstract; detailed hardware test logs were not provided.
- An emergency switch demonstrates request handling but does not prove automatic identification of authorised vehicles.
- LoRa performance depends on the selected module, antenna, radio settings, obstacles, and local conditions.
- A miniature LED model does not establish safety for real traffic infrastructure.
- GPS tracking, traffic-density sensing, central monitoring, and automated route selection are optional future enhancements unless implemented and tested.
- Measured improvement in emergency travel time requires controlled experiments or field evaluation.

5.7 Chapter Summary

This chapter defines the expected demonstration behaviour and the metrics needed to evaluate the system. Numerical results should be inserted only after the prototype has been tested and logs have been collected.

CHAPTER 6 — APPLICATIONS AND DEMONSTRATION

6.1 Application Overview

A coordinated emergency-priority network may be useful in environments where emergency vehicles travel through multiple signalised junctions. The current project is a prototype concept and should be treated as a research and educational demonstration until it has undergone safety, reliability, and regulatory validation.

Application area	Potential role
Ambulance routes	Prepare successive junctions along a configured hospital route
Fire and rescue response	Support coordinated priority along a route to an incident
Campus / industrial roads	Demonstrate controlled priority on a small private-road test network
Smart-city research	Prototype a low-rate wireless junction coordination approach
Traffic-control education	Teach embedded control, wireless messages, state machines, and system testing

6.2 Demonstration Sequence

7. Power on each ESP32 junction node and confirm normal signal cycling.
8. Activate the emergency request at Junction 1.
9. Observe request validation and safe local signal priority.
10. Confirm that the LoRa message is transmitted and received by Junction 2.
11. Observe Junction 2 preparing the intended signal phase; repeat for Junction 3 if present.
12. Trigger the vehicle-passage or clear condition.
13. Confirm that each active node clears the request and restores normal traffic.
14. Repeat with a duplicate or invalid packet to verify message validation.

6.3 Demonstration Evidence to Include

For the final submission, include photographs of the assembled ESP32 and LoRa nodes, a wiring diagram, screenshots of Serial Monitor logs, and a table of measured test results. Add timestamps for request activation, packet transmission, reception, priority activation, and restoration. Any screenshots or measured values should come from the actual project implementation.

Illustrative Three-Junction Model Layout

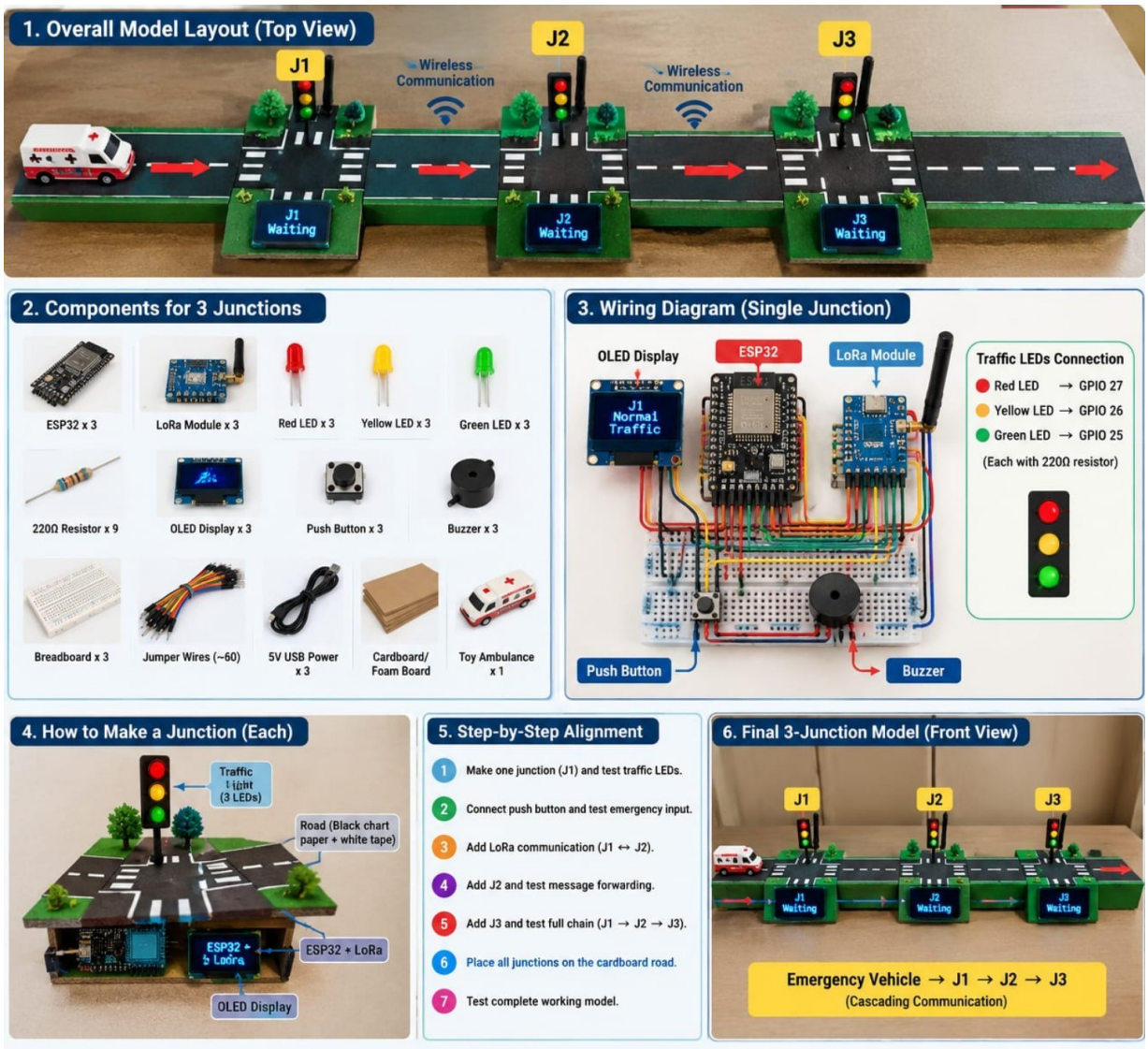


Figure 6.1: Overall model layout, components, single-junction wiring, and three-junction cascading demonstration.

6.4 Chapter Summary

This chapter outlined application areas and a repeatable demonstration sequence. The project should clearly distinguish features that have been demonstrated from those planned for future implementation.

CHAPTER 7 — CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Smart Emergency Vehicle Priority System proposes a low-cost method for coordinating emergency vehicle priority across consecutive traffic-signal junctions. ESP32 microcontrollers handle local request processing and signal control, while LoRa communication carries emergency information to upcoming junctions. The cascading design aims to reduce repeated stops by allowing downstream junctions to prepare in advance.

The system workflow includes request detection, message generation, wireless forwarding, safe signal priority, vehicle passage, and restoration of normal operation. The current report documents the architecture and a practical prototype methodology. Since numerical hardware results were not included in the supplied project materials, performance improvements remain to be verified through recorded experiments.

7.2 Achievement of Project Objectives and Status

Objective	Status in this report
ESP32-based emergency request handling	Design and implementation approach described
LoRa junction-to-junction communication	Architecture and message flow described; measured performance to be recorded
Local traffic-signal priority	Safe prototype state sequence proposed
Cascading preparation of upcoming junctions	Multi-node workflow described
Restoration of normal traffic	Clear/timeout behaviour specified for implementation
Performance evaluation	Metrics and test cases provided; numerical values pending

7.3 Future Enhancements

- GPS-based route tracking to select the next junction automatically.
- Automatic emergency vehicle detection using a secure vehicle-side transmitter or authorised beacon.
- Traffic-density sensors to support more context-aware phase preparation.
- Acknowledgements, retries, message authentication, and anti-replay protections.
- Central dashboard for monitoring active requests and junction status.
- Testing with more nodes and different radio conditions.
- Integration with certified traffic controllers only after engineering review and official approval.
- Controlled experiments to compare travel delay with and without cascading priority.

7.4 Final Summary

The project demonstrates the concept of extending emergency vehicle priority beyond one intersection by forwarding requests to downstream junctions. Its next important step is to build and test the multi-node prototype, document the wiring and firmware configuration, and report measured results without overstating what has been verified.

REFERENCES

[1] Espressif Systems, “ESP32 Series Datasheets and Technical Documentation,” Espressif Systems. Consult the exact board documentation used in the prototype.

[2] Semtech Corporation, “LoRa Modulation Basics,” technical application material. Consult the documentation corresponding to the selected radio module.

[3] Arduino, “Arduino IDE Documentation,” Arduino documentation portal.

[4] International Telecommunication Union (ITU), radio-frequency and spectrum recommendations applicable to the deployment region; verify the local rules before operation.

[5] Project source materials supplied by the authors: “Smart Emergency Vehicle Priority System” presentation and “Smart Emergency Vehicle Priority System” abstract.

[6] A. Ibraheem, A. Asiri, B. Asiri, and H. Alhefzi, “Intelligent Traffic Light Control System for Emergency Vehicles Using Deep Learning and Signal Synchronization,” IEEE Access, vol. 13, 2025, doi: 10.1109/ACCESS.2025.3644805.

Note: Verify the exact hardware documentation and access dates before final submission. The peer-reviewed IEEE Access article listed as reference [6] provides related background on intelligent emergency-vehicle traffic-signal control; its deep-learning detection approach differs from this project’s ESP32–LoRa cascading-junction prototype.

APPENDIX A — SAMPLE MESSAGE AND TEST LOG

APPENDIX B — CONFERENCE-PAPER IMPLEMENTATION AND TEST PROTOCOL

B.1 Junction-node algorithm

1. Receive the packet and verify its format and emergency status.
2. Check whether the junction belongs to the configured route and direction.
3. Reject malformed, stale, duplicate, or irrelevant requests.
4. Schedule a safe local priority transition after the required clearance interval.
5. Display or log request status and forward the event to the next configured junction.
6. Monitor simulated passage, cancellation, or timeout.
7. Clear the event and restore normal signal operation.

B.2 Expected behaviour for fault cases

Test case	Expected observation
Duplicate packet	No second priority cycle starts for the same event
Malformed packet	Packet is rejected and signal state does not change
Downstream node unavailable	Failure/timeout is logged; delivery is not falsely reported
Cancellation / timeout	Request is cleared and normal signal sequence resumes
Normal completion	Every active junction clears the event after simulated passage

A.1 Illustrative Message

The following is a human-readable example only; it is not a required wire format.

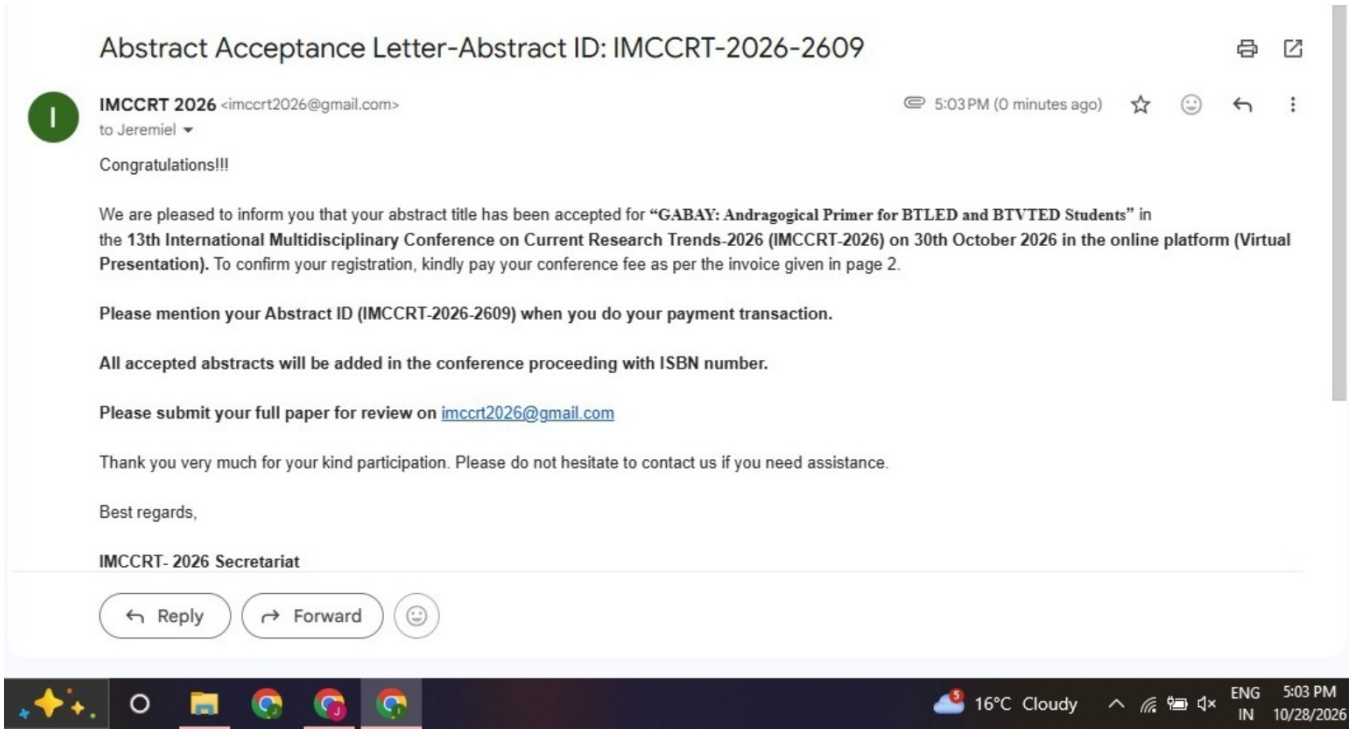
```
message_id=EV-0007; vehicle_type=AMBULANCE; emergency_status=ACTIVE;
current_junction=J1; next_junction=J2; direction=NORTH; priority=HIGH;
sequence_number=12
```

A.2 Suggested Test Log

Timestamp	Event	Junction	Message ID	Result / Notes
_____	Request activated	J1	_____	_____
_____	Priority state entered	J1	_____	_____
_____	Packet transmitted	J1	_____	_____
_____	Packet received	J2	_____	_____
_____	Priority prepared	J2	_____	_____
_____	Clear / passage event	J1/J2	_____	_____
_____	Normal state restored	J1/J2	_____	_____

A.3 Information to Complete Before Submission

- Confirm the official project title and degree/department wording required by the department.
- Confirm student names and register numbers on the cover page.
- Confirm the guide and Head of Department names/designations.
- Insert the actual circuit diagram and photographs of the prototype.
- Record the ESP32 board model, LoRa module model, radio frequency configuration, and GPIO pin assignments.
- Fill all evaluation tables using measured results from repeated tests.
- Verify references and format according to the department's current report guidelines.


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